

TrES-5b

R-1957

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_190907 · created 2026-08-06T16:11:28+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458733.8439 +/- 0.0029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.009 +/- 0.051
Transit depth (Rp/Rs)^2	0.009 +/- 0.097 [%]
Orbital Inclination (inc)	79.7 +/- 2.92
Airmass coefficient 1 (a1)	0.68 +/- 0.56
Airmass coefficient 2 (a2)	0.3 +/- 0.49
Scatter in the residuals of the lightcurve fit is	85.9 %
Best Comparison Star	None
Optimal Aperture	0.94
Optimal Annulus	7.8
Transit Duration (day)	0.019 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.009 $\pm$ 0.051 vs 0.142 (deviation 1.06 σ)
Duration fit vs published	0.019 d vs 0.0758 d (deviation 0.92 σ)
Mid-time O-C	4.6 min
Depth SNR	0.1
Residual scatter	85.9 %

Light curve

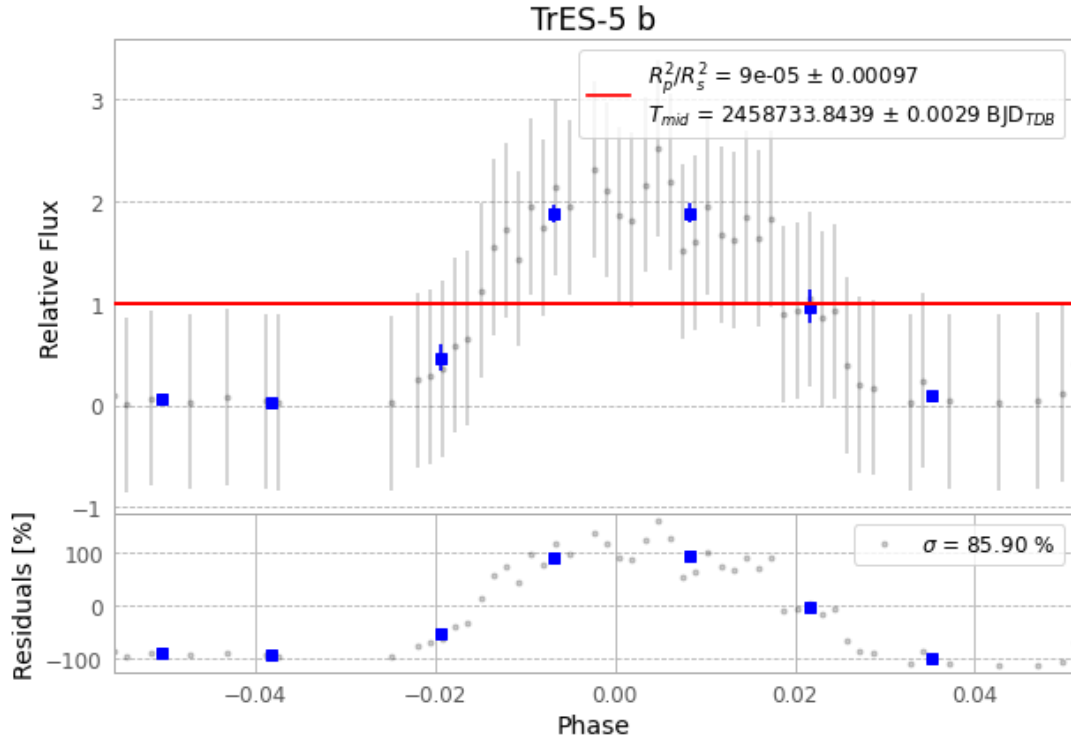


Figure 1 — FinalLightCurve_TrES-5 b_2019-09-06.png (SHA-256 a05b8c87e550dfbb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [447, 278], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ee9160d0b37b64e1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

77 FITS frames (51 MB), TRES-5190907061212.FITS → TRES-5190907100012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-190907.log (55d8bee73686...), FinalLightCurve_TrES-5 b_2019-09-06.png (a05b8c87e550...), FinalLightCurve_TrES-5 b_2019-09-06.csv (ad445dc6cb39...)

Compute resources

Wall time 249.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 16:11Z.